

Name: _____

Date: _____

How Volcanoes Form

Grade 6 reading comprehension · Informational text · 14 points · About 30 minutes

Inside Earth

Earth's outer layer, called the crust, is not one solid piece. It is broken into giant slabs called tectonic plates, which slowly float on the hot, partly melted rock of the mantle beneath. The plates move only an inch or two each year, about as fast as your fingernails grow, but over millions of years that is enough to reshape continents. The edges of the plates are the most active places on Earth, where mountains rise and the ground shakes.

Where volcanoes form

Most volcanoes form where two plates meet. In one common case, a heavy ocean plate slides beneath another plate in a process called subduction. As the plate sinks deeper, water trapped in its rocks is squeezed out, which helps nearby rock melt. The melted rock, called magma, is lighter than the rock around it, so it slowly rises. Other volcanoes form over hot spots, places where very hot material from deep inside Earth melts through the crust, even far from the edge of a plate. The Hawaiian Islands formed this way: as the Pacific Plate slowly moved over a hot spot, one volcano after another rose from the sea.

Inside a volcano

Magma collects underground in a huge pocket called a magma chamber. When the pressure builds enough, the magma forces its way up through an opening called a vent and erupts. Once magma reaches the surface, it is called lava. Some volcanoes erupt gently, with lava flowing like a slow river. Others erupt explosively, blasting ash high into the air, depending on how thick the magma is and how much gas it holds. Many volcanoes have a bowl-shaped dip called a crater around the main vent.

The Ring of Fire

About 75 percent of Earth's volcanoes lie in a huge horseshoe around the Pacific Ocean known as the Ring of Fire. Many of Earth's earthquakes also happen there, because both are caused by the same moving plates. Scientists called volcanologists study the Ring of Fire closely, hoping to warn people before an eruption.

1. What are tectonic plates? (1 point)

2. What does the word “subduction” mean? Cite the sentence that helped you work it out. (2 points)

3. Quote the sentence that explains why magma rises. Use quotation marks. (2 points)

4. The text says plates move about as fast as fingernails grow, yet they can reshape continents. What does this suggest about how long big changes in Earth take? Cite evidence. (2 points)

5. How are volcanoes that form over hot spots different from volcanoes at plate edges? Cite evidence from the text. (2 points)

6. Number these events 1 to 4 in the order they happen. (2 points)

- ☐ Magma forces its way up through a vent.
- ☐ Magma collects in a magma chamber.
- ☐ Rock melts to form magma.
- ☐ Magma rises toward the surface.

7. Why do you think earthquakes and volcanoes both happen in the Ring of Fire? Cite evidence. (2 points)

8. Choose the statement that best sums up the central idea of the text. Write its letter. (1 point)

- A. Volcanoes only erupt under the ocean, and they never form anywhere on land or on islands.
- B. Moving plates and rising magma form most volcanoes, especially around the Pacific Ocean.
- C. The crater is the most important part of a volcano, because every eruption starts there.
- D. Earthquakes are harmless, so scientists do not need to study the Ring of Fire.

Answer: _____

Writing task: Write a news report about a volcano erupting. Include what scientists know about why it happened.
